"PERFORMANCE OF MUSTARD UNDER DIFFERENT PRUNING REGIMES OF NEEM IN AGROFORESTRY"

Anshuman Singh

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PROJECT WORK & DISSERTATION

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RANI LAKSHMI BAI CENTRAL AGRICULTURAL UNIVERSITY
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CERTIFICATE

This is to certify that ***Mr. Anshuman Singh***, Id No. ***F/070/20***, has submitted the project report of the work presented in the **Project Work & Dissertation(FPW 401)** entitled "***PERFORMANCE OF MUSTARD UNDER DIFFERENT PRUNING REGIMES OF NEEM IN AGROFORESTRY***" in partial fulfilment of the requirement for the award of degree of B. Sc. (Hons) Forestry of Rani Lakshmi Bai Central Agricultural University, Jhansi-284003, which is an authentic work carried out under our supervision and guidance.

To the best of our knowledge, the content of this thesis does not form a basis for the award of any previous degree to anyone else.

Date:

Dr. Ram Prakash Yadav

Associate Professor, Forestry

Rani Lakshmi Bai Central Agricultural University, Jhansi-284003

Mr. Suresh Ramanan

Scientist, Agroforestry

ICAR-CAFRI, Jhansi

Head of the Department

Silviculture & Agroforestry
Rani Lakshmi Bai Central

Agricultural University

Jhansi-284003

Dean

College of Horticulture & Forestry
Rani Lakshmi Bai Central

Agricultural University

Jhansi-284003

DECLARATION

This is to certify that the whole of the research work reported here in the project work for the partial fulfilment of the requirements for the degree of ***Bachelor of Science (Hons) Forestry*** is the result of investigation done by me, under the direct guidance and supervision of ***Dr. Ram Prakash* Yadav**, Associate Professor (Forestry), College of Horticulture and Forestry, Rani Lakshmi Bai Central Agricultural University, Jhansi, and no part of the work has been submitted for any other degree so far.

Place: *Jhansi*

Date: ***Anshuman Singh***

Countersigned by:

Dr. Ram Prakash Yadav
Associate Professor
Department Silviculture & Agroforestry
College of Horticulture and Forestry
Rani Lakshmi Bai Central Agricultural University
Jhansi-284003, Uttar Pradesh

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Abstract

Neem-based agroforestry has emerged as one of the most promising agricultural practices that combines the beneficial attributes of neem trees with conventional farming practices. This project report focuses on the intercropping of mustard (*Brassica juncea*) under Agri-silvicultural systems with Neem (*Azadirachta indica*) and explores its potential advantages and challenges through in-depth examinations of growth parameters, yield, biometric and carbon estimation.

The growth analysis of mustard plants revealed significant results in their performance based on different treatments. While Sole mustard (T8) remained the best performer out of all, the treatment 60% Pruning + Apical cutting (T6) had great potential as it often performed at par.

The height (239.8 cm), number of primary branches (7.8), number of secondary branches (17.3), dry matter (78.1 g), number of pods per plant (258), number of seeds/pod (16.3) and average pod weight (0.203 g) was significantly higher in sole mustard (T8) at harvest.

The yield analysis further indicated variations in biological yield and straw yield, with sole mustard (T8) showing the highest values. Grain yield on the other hand was significantly higher in sole mustard (1444 kg/ha) compared to T1, T2 & T7, while it was at par with T3, T4, T5 & T6.

The tallest trees (5.12 m) with longest crown length (3.56 m) were seen in T7 (without pruning) which was significantly higher. The highest girth (42.6 cm), diameter (0.137 m), volume (0.063 m^3), above ground biomass (17.6 t/ha), below ground biomass (9.1 t/ha), total biomass (26.7 t/ha), above ground carbon (8.8 t/ha), below ground carbon (4.5 t/ha) and total carbon (13.3 t/ha) was significantly higher in 60% pruning + apical cutting (T6). These findings contribute to the understanding of agroforestry management and the potential impact of pruning and apical cutting on neem tree growth.

KEYWORDS: Agroforestry, Pruning, Apical, Biological yield, Grain yield, Straw yield, Harvest Index, Volume, Biomass, Carbon.

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1. INTRODUCTION

Agroforestry, the integrated practice of combining trees with crops or livestock on the same unit of land, has garnered significant attention as a sustainable land-use system. This approach not only contributes to food security and agricultural productivity but also offers numerous environmental benefits, including improved soil fertility, carbon sequestration, and biodiversity conservation (Nair, 1993). Among the various agroforestry practices, the integration of tree species with agricultural crops, commonly known as agri-silvicultural systems, has demonstrated considerable potential for enhancing crop yields and promoting ecological resilience. Intercropping, as an agricultural practice, involves the simultaneous cultivation of two or more crops within the same field. It offers several advantages, such as maximizing land use efficiency, enhancing resource utilization, and promoting ecological balance.

One such agri-silvicultural system involves the cultivation of mustard (*Brassica juncea*), a widely grown oilseed crop, alongside neem (*Azadirachta indica)* trees. Neem, a multipurpose tree species, has been extensively used in agroforestry systems due to its numerous benefits, including soil improvement, nitrogen fixation, and pest management properties (Panwar et al., 2011). However, the successful implementation of neem-based agroforestry systems requires careful management practices, particularly about tree pruning regimes.

Pruning, the selective removal of branches or shoots, is a crucial aspect of agroforestry management as it influences the growth, productivity, and resource allocation of both trees and associated crops (Nguyen et al., 2015; Wajja-Musukwe et al., 2008). Appropriate pruning strategies can optimize light interception, nutrient cycling, and resource partitioning, ultimately enhancing the overall productivity of the agroforestry system. Nevertheless, the impact of pruning practices on the performance of mustard under neem-based agroforestry systems remains largely unexplored.

Neem-based Agroforestry, which integrates neem trees into agricultural systems, has gained attention as an innovative approach that combines the benefits of both trees and crops. This project report focuses on the intercropping of mustard (*Brassica juncea*) under neem-based agroforestry and explores its potential advantages and challenges.

Mustard is a widely cultivated oilseed crop known for its high oil content and nutritional value. By intercropping mustard with neem trees, farmers can take advantage of the synergistic effects and ecological services provided by the trees. Neem trees offer numerous benefits, including

pest control, nutrient cycling, soil improvement, and microclimate modification. These advantages can positively influence the growth, yield, and quality of mustard crops, leading to improved overall agricultural productivity.

The primary objective of this project is to assess the performance of mustard intercropped under neem-based agroforestry and the effect of pruning on neem trees on the growth of mustard. By comparing the growth parameters & yield components of mustard in intercropped systems with those in monoculture systems and comparing it among the different treatments, we can evaluate the potential benefits of neem-based agroforestry. Additionally, the project aims to identify any challenges or trade-offs associated with this farming system and provide recommendations for successful implementation.

The findings of this study will contribute to the existing knowledge on neem-based agroforestry and intercropping systems. By evaluating the performance of mustard in the presence of neem trees, farmers and researchers can make informed decisions regarding crop selection, planting patterns, and management practices. Furthermore, this project will shed light on the ecological and economic sustainability of neem-based intercropping systems, providing valuable insights for agricultural stakeholders.

This report presents a comprehensive analysis of the intercropping of mustard under different pruning regimes of neem in agroforestry, incorporating experimental data, statistical analysis, and literature review. Through an in-depth examination of the growth, yield, and quality parameters, we aim to unravel the potential benefits and challenges associated with this innovative farming practice. The outcomes of this project will assist farmers, policymakers, and researchers in adopting and promoting sustainable agricultural practices that enhance productivity while preserving the environment.

Mustard (*Brassica juncea* L.) is one of the important oilseed crops of the world. Plant origin oil constitute important component of human diet, ranking third after cereals and animal products and are nutritionally superior to animal oil (Alam et al., 2015; Buttar and Aulakh, 1999, FAO, 2007). The productivity is quite lower than developed countries mainly due to sub-optimal application of fertilizers and cultivation on marginal lands in rainfed conditions (Kumawat et al., 2014). Variety selection is the most important decision to achieve higher crop yield by improving the fertilizer use efficiency and water use efficiency) Thus, growers' wisdom to select high-yielding varieties with better agronomic traits that match the ever-changing stresses in field conditions. It is also a fact that specified genotypes do not exhibit the same phenotypic characteristics in all environmental conditions. Thus, improved cultivar

selection is an important tool for improving productivity. Hot arid & semi-arid regions soils are thirsty and hungry and suffers from various atmospheric stresses (Bal and Minhas, 2017). Mustard is a crop of tropical as well as temperate zones requiring somewhat cool and dry weather for satisfactory growth and development. High temperature at flowering stage causes reduction in seed yield as it may lead to pollen sterility (Singh et al., 2014).

Standard procedures and observations were recorded on growth, yield attributes and yield. Plant height at different stages of crop growth were recorded for two randomly selected plants from sampling rows and measured from the ground to the tip of the plant by meter scale and averaged them in centimetres. Two plants per plot were cut from the ground level from sampling row at harvest stage. Those plants were first air-dried for 2-3 days following by oven dried at 60-65°C for 48 hours and dry weight was recorded in grams.

The cultivation of mustard (*Brassica juncea*) plays a vital role in the agricultural sector, serving as a crucial source of edible oil and contributing significantly to the economy. As the demand for sustainable agricultural practices continues to rise, researchers and farmers are exploring innovative techniques to enhance crop productivity while minimizing the negative impacts on the environment. Neem-based agroforestry has emerged as one such promising approach that combines the beneficial attributes of neem trees with conventional farming practices.

Neem (*Azadirachta indica*), a versatile tree native to the Indian subcontinent, possesses numerous agronomic and ecological benefits. Its unique properties, including pest control, nutrient cycling, and soil improvement, make it an ideal component for agroforestry systems. Neem-based agroforestry involves the deliberate integration of neem trees with crop cultivation, creating a mutually beneficial relationship between the tree species and agricultural crops. Neem exhibits multifarious uses with medicinal properties being its characteristic feature across generations. During the period of study, the dimensions of the trees were carefully measured along with the crown spread of the trees. The biomass and the carbon content of the stand was also calculated. The pruning of the trees, laterally & apical, showed different growth behaviour in the respective trees that sheds a light on the effect of tending in the form of pruning on neem and the effect on mustard crops in between.

Pruning is an important tending activity in forestry that involves the selective removal of branches, stems, or shoots from trees within a forest stand or plantation. It is typically carried out to improve the quality and value of the remaining trees for timber production. By pruning lower branches, foresters can reduce the formation of knots in the wood, which enhances the

overall grade and value of the lumber. Pruning also allows for better air circulation and light penetration within the stand, promoting the growth and development of the desired crop trees. Additionally, it can be used as a tool for stand management, where competing or interfering stems are removed to favour the growth of the most desirable individuals. When done correctly and at the appropriate times, pruning is an essential practice for optimizing the productivity and economic returns of managed forest resources.

The performance evaluation of mustard under neem-based agroforestry becomes an intriguing area of study due to the potential synergistic effects that may arise from this combination. This project report aims to investigate the impact of neem-based agroforestry on the growth, yield, and quality of mustard crops whilst also assessing the potential effects of the pruning treatments on the mustard crops. By examining various parameters such as plant height, no. of branches, no. of pods/plant, no. of seeds/pod, dry matter, etc. we can gain valuable insights into the overall performance of mustard in this innovative farming system.
Understanding the performance of mustard under neem-based agroforestry holds immense significance for sustainable agriculture. It provides an opportunity to explore alternative farming practices that promote ecological balance, reduce chemical inputs, and improve crop resilience. The results obtained from this study can guide farmers, policymakers, and researchers in making informed decisions regarding the adoption and promotion of neem-based agroforestry systems.

This report presents a comprehensive analysis of the performance of mustard under different pruning regimes of neem in agroforestry incorporating experimental data, statistical analysis, and relevant literature review. By delving into the intricate relationship between neem trees and mustard cultivation, we aim to unravel the potential benefits, challenges, and recommendations associated with this innovative farming approach. Ultimately, this project seeks to contribute to the advancement of sustainable agricultural practices and the overall well-being of farming communities.

- OBJECTIVES:
  i. Growth and dry matter accumulation of mustard before harvest and at harvest
  ii. Yield and yield attributes of mustard
  iii. Biometric analysis and carbon estimation of neem

2. REVIEW OF LITERATURE

Kumar et al. (2004), studied about the Effect of neem trees on yield of mustard varieties. This study clearly reveals that when we talk of growing mustard under the tree canopy condition, variety Varuna will be obviously our choice because with this variety we are going to harvest more. One of the desired advantages with this variety was that it showed consistent performance over the years. The other varieties which showed good performance were Rajat and Kranti.

Chakravarty et al. (2010), studied about the multipurpose utility of the tree borne oil seed species (TBOS) that makes them desirable species for agroforestry systems. Establishment of agroforestry systems with TBOS on vast areas of barren or scrub land and less fertile farmland will change the rural landscape and bring with it many environmental benefits. Wastelands belonging to farmers that can be brought under TBOS are available in plenty. The present production levels of some marginal farmlands are so low that farmers will be happy to try out alternative crops.

Newaj et al. (2012), conducted field experiments during 2006–08 at Jhansi to elucidate the influence of canopy pruning in white siris [*Albizia procera* (Roxb.) Benth.] on light interception and grain yield of intercrop. The experiment was conducted in randomized block design with three replications. Blackgram (*Vigna mungo* (L.) Hepper) - mustard (*Brassica juncea* L.Czern. and Coss.) crop sequence was taken as intercrop. The pruning of tree canopy was done with 3 pruning regimes namely 70% canopy pruning, 50% canopy pruning and control (unpruned tree). The results reveal that the growth of *A.procera* (dbh, height and canopy diameter) was significantly (P= 0.05) higher in control (unpruned trees) compared with 50% canopy pruning and 70% canopy pruning. Grain yield of black gram was two times higher in 70% canopy pruning than the yield obtained with unpruned trees. Similarly, the yield of mustard was three times higher in 70% canopy pruning than unpruned trees during both the years. The grain yield of intercrop under 50% canopy pruning was also significantly higher than unpruned trees. Light interception by intercrops was higher under 70% canopy pruning and lower with unpruned trees. The light interception increased with increasing distances from tree base. The grain yield of intercrops was linearly correlated with light interception.

Verma et al. (2017) studied about A paradigm shift in agroforestry practices in Uttar Pradesh. In UP, agroforestry practices vary according to different agro-climatic zones, land capability and socio-economic status of farmers. The variation is reflected in terms of diversity in agroforestry practices, and comparative advantage prompted a renewed interest to harness the vivid potential. Drawing on the representative literature, we have reviewed the status and pattern of tree–crop combinations of agroforestry practices across various regions of the state as well as productivity under different agroforestry systems, which shows traditional agriculture transforming to multifunctional agroforestry in UP.

Bijalwan et al. (2019) studied about agroforestry and extent of agriculture, horticulture, and agroforestry in MP. Information on diverse tree species under agroforestry systems, a mosaic of tree–crop combinations and horticulture species preferred by the farmers in different agro-climatic zones of the state. The authors have also tried to enlist the efforts of various government initiatives and other agencies involved in the extension of agroforestry in the state.

The roles of MP Forest Department, MP Forest Development Corporation, and allied Departments of the state in agroforestry and tree planting have also been discussed.

Sharma & Sah (2020), studied about the Comparative growth, yield, and yield attributes of the wheat under poplar and Eucalyptus based agroforestry system. The observations on growth and yield parameters of wheat crop were recorded during rabi season. Germination count, plant height, number of tillers and dry matter accumulation were higher under open farming system than agroforestry system (poplar and Eucalyptus). Among wheat varieties UP-2526 showed higher germination count and maximum height, number of tillers and dry matter accumulation. Yield attributes like grain-1 yield was higher in open farming (3.55t ha) system as compared to agroforestry system. Among wheat varieties grain-1 yield was significantly higher in UP-2628 (3.13t ha).

Shah et al. (2021), studied about the Yield performance & economic evaluation of mustard varieties under mango based Agri-horticulture practice in semi-arid tropics. This analysis conclude that the highest yielding variety was Arpan among mustard varieties. Economic evaluation we find it the cost of cultivation of Agri-Horticulture practice is higher than Sole-Horticulture practice, the gross monitoring return & net monitoring return were highest found in Arpan (Rs 62,153.85 ha−1 & Rs 36,916.45 ha−1 respectively) variety. The B/C ratio was not significantly difference but highest found in Arpan 0.966 in 1st year of plantation.

Das et al. (2022) studied about the Fruit based agroforestry systems -potential means for sustaining carbon sequestration, improving soil health and diet of community in red and lateritic zone of West Bengal, India. Total carbon stock as well as potential food energy production were higher in AFS than either in sole silvi tree, fruit tree or plots under alley crops, because of better performances of all tree and crop components in AFS. Mango with eucalyptus-based AFS gave higher carbon stock (62.33 t ha-1 yr-1) including improvement in degraded acidic soil pH (6.20%), SOC (11.86%), available N (9.09%), available P (13.97%), available K (11.64%) contents in soil. In that way, fruit-based agroforestry systems can be used as a viable alternative land use to improve soil health, diet, and livelihood security even of resource-poor farm families throughout the year, especially during their impoverished needy times in a year**.**

Kumar et al. (2022), studied about the Growth and yield performance of black gram (*Vigna mungo* L.) under malabar neem (*Melia dubia*) plantations in western zone of Tamil Nadu. The results of pooled analysis of two season shown that, treatment S0V2 recorded plant height (49.96 cm), number of branches (3.76), number of leaves (259.75), number of flowers (53.44), number of pods (22.85), seed yield per plant (5.38 g) and seed yield per hectare (0.82 tons) in open condition. Similarly in intercropping the growth and yield attributes of black gram were minimum performance recorded in CO 6 at closer spacing (3 × 1.5 m). In tree growth maximum pooled height increment (1.57 m) was observed in closer spacing (3 × 1.5 m) and maximum diameter recorded in wider spacing of both varieties. Hence wider spacing of S3 (4 × 4 m) can be suggested for intercropping under *M. dubia* plantations up to 4 years.

Kaushik et al. (2023) studied about the *Ailanthus excelsa* L. based agroforestry systems in North-West Indian dry lands: effect on yields of intercrops and soil properties. In the present study, yields of agricultural crops were not influenced by *A. excesla*, irrespective of crop or tree density; however, higher yields were recorded under sole cropping in both the cropping systems. Among different tree densities, higher crop yields were recorded with 50 trees ha^{-1}. Soil fertility status in terms of organic matter (OM), available P and K improved under agroforestry systems as compared to sole cropping. Growth of *A. excelsa* under tree densities and in association with crops was at par, maximum girth (47.10 cm) at breast height was recorded under density of 200 trees ha−1 with cluster bean-wheat cropping sequence. Two years experimentation showed maximum benefits under agroforestry with 200 trees ha^{-1}.

Mehta & Kaushik (2023), studied about the Impact of planting geometry on *Ailanthus excelsa* L. based silvo-arable systems for food and biomass production. *Ailanthus excelsa* Roxb. (Mahaneem) is one of the promising fast-growing and multipurpose trees of dry areas and used in various plantation programs due to its ability to grow well with less rainfall. Therefore, in this study, *Ailanthus excelsa* based agroforestry systems (growing of trees and food crops simultaneously on the same piece of land) with different plant geometry, i.e., 10 × 20, 10 × 10, 10 × 6.5 and 10 × 5 m (50, 100, 150 and 200 trees ha^{-1}) were assessed for food and biomass production. Among all the plant geometries, the plant geometry of 10 × 20 m exhibited higher yield components and yield viz. number of effective tillers and siliquae per plant (306.7 m−2 and 268.42), number of grains per spike and per siliqua (44.83 and 12.5), test weight (40.21 and 5.89 g), grain and seed yield (3.93 and 1.90 t ha^{-1}), straw and stover yield (5.0 and 2.70 t ha^{-1}); and root weight (2.11 and 0.76 t ha^{-1}) and total biomass (11.04 and 5.36 t ha^{-1}) of wheat and Indian mustard, respectively. Biomass of 28.6 t $ha^{-1}$from trees and 5.9 t $ha^{-1}$from wheat (only straw and underground) under 5 × 4 m tree geometry can be potential feedstock for biofuel production on sustainable basis.

3. MATERIALS AND METHODS

3.1. Experimental area

The experiment was conducted in neem plantation at Forestry research farm, Bhojla, RLBCAU, Jhansi, Uttar Pradesh. The experiment location situated at 25.44° N latitude and 78.57°E longitude at an altitude of 300 meters above mean sea level.

3.2. Climate

Bundelkhand is a hot and semi-humid region. Minimum temperature varies from around 6°C in Chhatarpur to 12°C in Sagar. Maximum temperature varies from 38°C in Sagar to 43°C in Banda. Generally, the hottest days are in May and the coldest days are in December or January.

3.3. Experimental Details

- **Crop:** Mustard (*Brassica juncea* L.)
- **Variety of Crop:** Giriraj
- **Tree:** Neem (*Azadirachta indica* L.)
- **Number of treatments:** Eight
- **Treatment details:**

Symbol	Treatment
T1	20% Pruning
T2	20% Pruning + Apical cutting
T3	40% Pruning
T4	40% Pruning + Apical cutting
T5	60% Pruning
T6	60% Pruning + Apical cutting
T7 (Control)	Without pruning
T8 (Open)	Sole mustard

- **Replications:** Three
- **Total plots under study:** Twenty-four plots of neem and mustard based agroforestry system under different treatments.
- **Design:** Randomized Block Design (RBD)

- **Spacing in *Azadirachta Indica:*** 6 m × 5 m
- **Plot size (Mustard):** 5 m x 5 m
- **Cultivation practices:** Recommended package and practices have been followed for growing of mustard crop.

3.4. Observations Recorded

The data for growth analysis was taken by tagging two plants per plot and observing them. The height of mustard plants was recorded using metre scale. number of primary branches and secondary branches and no. of pods/plant were counted manually. Ten pods were selected to measure the pod attributes (pod length, pod weight & seeds/pod). After completing all these, the mustard plants were dried to take the dry weight, so all those plants were packed according to its specific replication and treatment then dried in hot air oven for 48 hours or more.

For yield analysis, the average dry weight of plots (Biological yield) was collected from each plot and converted into kg/ha values. Threshed grains per plot were weighed by weighing balance, the average grain weight was converted into kg/ha basis (Grain yield). Deducting the value of grain yield values from biological yield values, straw yield values were also calculated.

The biometric analysis and carbon estimation has been done for all the trees present at the project site. The biometric measurements and carbon estimation of neem were done by using clinometer, measuring tapes and linear regression equations derived from well researched projects of areas having similar conditions to the conditions present at the project site.

All the calculations were done at ease with the help of ***Microsoft excel***.

3.4.1. Growth analysis of Mustard

i. Height (cm)
ii. Number of primary branches
iii. Number of secondary branches
iv. Dry matter (g)

3.4.2. Yield attributes for Mustard

i. Average number of pods/plants
ii. Average number of seeds/pods
iii. Average pod length (cm)
iv. Average pod weight (g)

3.4.3. Yield analysis

i. Biological yield (kg/ha)
ii. Grain yield (kg/ha)
iii. Straw Yield (kg/ha)
iv. Grain to Straw Ratio
v. Harvest Index

3.4.4. Biometric analysis of Neem

i. Tree height (m)
ii. Bole height (m)
iii. Crown length (m)
iv. Crown Spread (m^2)
v. Girth at Breast Height (cm)
vi. Diameter at Breast Height (m)
vii. Volume (m^3)

3.4.5. Carbon estimation of Neem

i. Above ground Biomass (t/ha)
ii. Below ground Biomass (t/ha)
iii. Total Biomass (t/ha)
iv. Above ground Carbon stock (t/ha)
v. Below ground Carbon stock (t/ha)
vi. Total Carbon stock (t/ha)

3.5. Statistical analysis

The data collected was compiled and analysed for working out the correlations following the procedure described by the method of analysis of variance and significance done using factorial completely randomized block design on the software ***OPSTAT*** made by CCS HAU Hisar, Haryana.

Pic 1. Experimental site

Pic 2. Growth analysis of mustard

Pic 3. Harvesting of mustard

Pic 4. Biological yield estimation

Pic 5. Grain yield

Pic 6. Dry weight estimation

Pic 7. Yield attributes of mustard

Pic 8. Biometric analysis of neem

4. RESULTS & DISCUSSIONS

4.1. GROWTH ANALYSIS OF MUSTARD

4.1.1. Before Harvest

The height, number of primary branches and number of secondary branches was significantly higher in ***Sole Mustard (T8)***.

The mustard plants were the tallest in the plots ***Sole mustard (T8***) with the height being ***189.9 cm*** which was significantly better than the plots treated with ***T1***, ***T2***, ***T3***, ***T4***, ***T5***, ***T6*** & ***T7***.

The no. of primary branches was the most in ***Sole mustard (T8)*** with ***5.8 branches*** which was significantly better than the plots treated with ***T1***, ***T2*** & ***T7*** while it was at par with the plots treated with ***T3***, ***T4***, ***T5***, ***T6*** & ***T8***.

The no. of secondary branches was the highest in mustard plants of the plots with ***Sole mustard (T8***) with ***15.7 branches*** which was significantly better than the plots treated with ***T1***, ***T2***, ***T3*** & ***T7*** while it was at par with the plots treated with ***T4***, ***T5*** & ***T6***.

Table 1. Growth analysis of mustard before harvest (Height, No. of primary branches & No. of secondary branches) under Neem based agroforestry

Treatments	Height (cm)	No. of Primary Branches	No. of Secondary Branches
20% Pruning (T1)	162.2	4.0	12.7
20% Pruning + Apical cutting (T2)	170.4	4.8	12.8
40% Pruning (T3)	169.9	5.0	13.5
40% Pruning + Apical cutting (T4)	175.5	5.0	14.8
60% Pruning (T5)	185.8	5.3	15.0
60% Pruning + Apical cutting (T6)	184.8	5.7	15.3
Without pruning (T7)	153.6	3.2	12.5
Sole mustard (T8)	189.9	5.8	15.7
C.D.	2.61	0.91	1.87
SE(m)	0.85	0.29	0.61

4.1.2. At Harvest

The height, number of primary branches, number of secondary branches and dry matter was significantly higher in ***Sole mustard (T8)***.

The mustard plants were the tallest in ***Sole mustard (T8)*** with the height being ***239.8 cm*** which was significantly better than ***T1***, ***T2***, ***T3***, ***T4***, ***T5*** & ***T7*** and at par with ***T6***.

The no. of primary branches was highest in ***Sole mustard (T8)*** with ***7.83 branches*** which was significantly better than the plots treated with ***T1***, ***T2***, ***T3***, ***T4*** & ***T7*** while it was at par with the plots treated with ***T5*** & ***T6***.

The no. of secondary branches was also highest in ***Sole mustard (T8)*** with ***17.3 branches*** which was significantly better than the plots treated with ***T1***, ***T3***, ***T4***, ***T6*** & ***T7*** while it was at par with the plots treated with ***T4***, ***T5*** & ***T6***.

The highest dry matter was seen in ***Sole mustard (T8)*** at ***78.1 g*** which was significantly better than mustard under neem-based agroforestry.

Table 2. Growth analysis of mustard at harvest (Height, No. of primary branches, No. of secondary branches & Dry matter) under Neem based agroforestry

Treatments	Height (cm)	No. of Primary Branches	No. of Secondary Branches	Dry Weight (g)
20% Pruning (T1)	190.5	5.8	14.0	43.4
20% Pruning + Apical cutting (T2)	208.4	6.0	14.5	44.4
40% Pruning (T3)	217.7	6.5	15.2	49.8
40% Pruning + Apical cutting (T4)	222.4	6.7	16.0	52.2
60% Pruning (T5)	227.8	7.0	16.5	58.8
60% Pruning + Apical cutting (T6)	234.3	7.3	17.0	66.7
Without pruning (T7)	180.9	4.8	12.5	42.8
Sole mustard (T8)	239.8	7.8	17.3	78.1
C.D.	5.76	1.01	1.40	3.57
SE(m)	1.88	0.33	0.46	1.17

4.2. YIELD ATTRIBUTES

Number of pods/plants, number of seeds/pod and average pod weight were significantly higher in ***Sole Mustard (T8)***.

The average no. of pods/plant was highest in ***Sole mustard (T8)*** with ***183 pods/plant*** (before harvest) and ***258 pods/plant*** (at harvest), which was significantly better than all other treatments. At harvest, the average no. of pods/plant was again highest in ***Sole mustard (T8)*** with ***258 pods/plant*** which was significantly better than all other treatments.

The average number of seeds per pod was the most in ***Sole mustard (T8)*** at ***16.3 seeds/pod*** which was significantly better than ***T7*** and was at par with ***T1***, ***T2***, ***T3***, ***T4***, ***T5*** & ***T6***.

The average weight of the mustard pods was highest in ***Sole mustard (T8)*** at ***0.203 g*** which was significantly better than all other treatments.

The effect of different treatments was non-significant for average length of pods. The average length of pods was numerically higher in ***60% Pruning + Apical cutting (T6)*** at ***5.43 cm*** and numerically lower in ***20% Pruning + Apical cutting (T2)*** at ***4.86 cm***.

Table 3. Yield attributes of mustard (No. of pods/plant, No. of seeds/pod, Average pod length, Average pod weight) under Neem based agroforestry

Treatments	**No. of Pods/Plant**		**No. of Seeds/Pod**	**Average Pod Length (cm)**	**Average Pod Weight (g)**
	BEFORE HARVEST	*AT HARVEST*			
20% Pruning (T1)	134	161	13.7	5.20	0.143
20% Pruning + Apical cutting (T2)	140	165	14.3	4.86	0.140
40% Pruning (T3)	141	176	14.7	4.92	0.153
40% Pruning + Apical cutting (T4)	149	193	15.0	5.12	0.147
60% Pruning (T5)	168	217	15.0	5.38	0.157
60% Pruning + Apical cutting (T6)	171	237	15.3	5.43	0.167
Without pruning (T7)	131	146	11.0	5.31	0.133
Sole mustard (T8)	183	258	16.3	5.05	0.203
C.D.	7.68	19.11	2.76	NS	0.024
SE(m)	2.510	6.240	0.903	0.231	0.008

4.3. YIELD ANALYSIS

The effect of different treatments was non-significant for biological yield, straw yield, grain to straw ratio and harvest index.

Biological yield in ***Sole mustard (T8)*** was numerically higher at ***4573 kg/ha*** and numerically lower in the plots ***Without pruning (T7***) at ***4312 kg/ha***.

Straw yield in ***60% Pruning + Apical cutting (T6)*** plots was numerically higher at ***3153 kg/ha*** and numerically lower in the plots ***Without pruning (T7)*** with ***2980 kg/ha***.

Grain to straw ratio in ***Sole mustard (T8)*** was numerically higher at ***0.461*** and numerically lower in ***20% Pruning + Apical cutting (T2)*** at ***0.444***.

Harvest index in ***Sole mustard (T8)*** was numerically higher at ***0.316*** and numerically lower in ***40% Pruning + Apical cutting (T4)*** at ***0.301***.

Grain yield on the other hand was highest in ***Sole mustard (T8)*** at ***1444 kg/ha*** which was significantly better than ***T1***, ***T2*** & ***T7*** and at par with ***T3***, ***T4***, ***T5*** & ***T6***.

Table 4. Yield analysis of mustard (Biological yield, Grain yield, Straw yield, Grain to Straw ratio & Harvest index) under Neem based agroforestry

Treatments	Biological Yield	Grain Yield	Straw Yield	Grain to Straw Ratio	Harvest Index
20% Pruning (T1)	4,353	1,357	2,996	0.452	0.312
20% Pruning + Apical cutting (T2)	4,391	1,352	3,039	0.444	0.308
40% Pruning (T3)	4,497	1,408	3,089	0.455	0.313
40% Pruning + Apical cutting (T4)	4,517	1,400	3,117	0.449	0.301
60% Pruning (T5)	4,539	1,404	3,135	0.447	0.309
60% Pruning + Apical cutting (T6)	4,567	1,414	3,153	0.448	0.310
Without pruning (T7)	4,312	1,332	2,980	0.446	0.309
Sole mustard (T8)	4,573	1,444	3,129	0.461	0.316
C.D.	NS	66.6	NS	NS	NS
SE(m)	92.295	21.7	67.934	0.003	0.001

4.4. BIOMETRIC ANALYSIS OF NEEM

The tallest trees were seen in the plots ***Without pruning (T7)*** with the height being ***5.12 m*** which was significantly better than ***T1***, ***T2***, ***T4***, ***T5*** & ***T6*** and at par with ***T3***.

In terms of bole height, the highest bole height is seen in the plots with ***40% Pruning (T3)*** with ***1.63 m*** which was significantly better than ***T2***, ***T5*** & ***T6*** and at par with ***T1***, ***T4*** & ***T7***.

The longest crown length was seen in the ***trees Without pruning (T7)*** with ***3.56 m*** which was significantly better than ***T1***, ***T2***, ***T4***, ***T5*** & ***T6*** and at par with ***T3***.

The largest crown spread was ***21.8 m^2*** seen in the trees treated with ***40% Pruning + Apical cutting (T4)*** which was significantly better than ***T1***, ***T2***, ***T5*** & ***T6*** and at par with ***T3*** & ***T7***.

Table 5. Biometric analysis of Neem (Tree height, Bole height, Crown length & Crown spread)

Treatment	Tree Height (m)	Bole Height (m)	Crown Length (m)	Crown Spread (m ^{2})
20% Pruning (T1)	4.70	1.54	3.16	13.5
20% Pruning + Apical cutting (T2)	4.35	1.46	2.89	13.7
40% Pruning (T3)	5.02	1.63	3.39	19.0
40% Pruning + Apical cutting (T4)	4.71	1.56	3.15	21.8
60% Pruning (T5)	4.52	1.51	3.01	15.8
60% Pruning + Apical cutting (T6)	4.47	1.37	3.10	9.3
Without pruning (T7)	5.12	1.56	3.56	18.7
C.D.	0.33	0.10	0.33	4.34
SE(m)	0.10	0.03	0.10	1.41

The girth at breast height, diameter at breast height and volume was significantly higher in ***60% Pruning + Apical cutting (T6).***

The girth at breast height was the highest in trees treated with ***60% Pruning + Apical cutting (T6)*** having ***42.6 cm*** girth which was significantly better than the ***T1***, ***T2*** & ***T4*** and at par with ***T3***, ***T5*** & ***T7***.

The diameter at breast height of the trees treated with ***60% Pruning + Apical cutting (T6)*** was the highest at ***0.137 m*** which was significantly better than ***T1***, ***T2*** & ***T5*** and at par with ***T3***, ***T4*** & ***T7***.

The volume was highest in the trees treated with ***60% Pruning + Apical cutting (T6)*** at ***0.063 m*** ^{3} which was significantly better than ***T1*** & ***T2*** and at par with ***T3***, ***T4***, ***T5*** & ***T7***.

Table 6. Biometric analysis of Neem (GBH, DBH & Volume)

Treatment	**GBH (cm)**	**DBH (m)**	**Volume (m^3)**
20% Pruning + Without Apical cutting (T1)	35.3	0.113	0.033
20% Pruning + Apical cutting (T2)	35.5	0.113	0.037
40% Pruning + Without apical cutting (T3)	39.6	0.123	0.050
40% Pruning + Apical cutting (T4)	39.5	0.123	0.053
60% Pruning + Without apical cutting (T5)	38.0	0.120	0.043
60% Pruning + Apical cutting (T6)	42.6	0.137	0.063
Without pruning (T7)	41.7	0.133	0.060
C.D.	4.443	0.014	0.020
SE(m)	1.426	0.005	0.006

4.5. CARBON ESTIMATION OF NEEM

The above ground biomass, below ground biomass, total biomass, above ground carbon, below ground carbon and total carbon was significantly higher in ***60% Pruning + Apical cutting (T6).***

The Above Ground Biomass was the highest in the trees treated with ***60% Pruning + Apical cutting (T6)*** at ***17.6 t/ha*** which was significantly better than ***T1***, ***T2*** & ***T5*** and at par with ***T3***, ***T4*** & ***T7***.

The Below Ground Biomass was the highest in the trees treated with ***60% Pruning + Apical cutting (T6)*** at ***9.1 t/ha*** which was significantly better than ***T1***, ***T2*** & ***T5*** and at par with ***T3***, ***T4*** & ***T7***.

The Total Biomass was the highest in the trees treated with ***60% Pruning + Apical cutting (T6)*** at ***26.7 t/ha*** which was significantly better than ***T1***, ***T2*** & ***T5*** and at par with ***T3***, ***T4*** & ***T7***.

The Above Ground Carbon was the highest in the trees treated with ***60% Pruning + Apical cutting (T6)*** at ***8.8 t/ha*** which was significantly better than ***T1***, ***T2*** & ***T5*** and at par with ***T3***, ***T4*** & ***T7***.

The Below Ground Carbon was the highest in the trees treated with ***60% Pruning + Apical cutting (T6)*** at ***4.5 t/ha*** which was significantly better than ***T1***, ***T2*** & ***T5*** and at par with ***T3***, ***T4*** & ***T7***.

The Total Carbon was the highest in the trees treated with ***60% Pruning + Apical cutting (T6)*** at ***13.3 t/ha*** which was significantly better than ***T1***, ***T2*** & ***T5*** and at par with ***T3***, ***T4*** & ***T7***.

Table 7. Carbon estimation of Neem (Above ground biomass, Below ground biomass, Total biomass, Above ground carbon, Below ground carbon & Total carbon)

Treatment	Above Ground Biomass (t/ha)	Below Ground Biomass (t/ha)	Total Biomass (t/ha)	Above Ground Carbon (t/ha)	Below Ground Carbon (t/ha)	Total Carbon (t/ha)
20% Pruning (T1)	11.7	5.7	17.4	5.8	2.8	8.7
20% Pruning + Apical cutting (T2)	11.9	5.8	17.8	5.9	2.9	8.9
40% Pruning (T3)	15.2	7.7	22.9	7.6	3.8	11.4
40% Pruning + Apical cutting (T4)	14.9	7.5	22.4	7.4	3.7	11.2
60% Pruning (T5)	13.7	6.8	20.5	6.8	3.4	10.2
60% Pruning + Apical cutting (T6)	17.6	9.1	26.7	8.8	4.5	13.3
Without pruning (T7)	16.7	8.6	25.3	8.4	4.3	12.6
C.D.	3.61	2.07	5.68	1.81	1.04	2.84
SE(m)	1.16	0.66	1.82	0.58	0.33	0.91

5. CONCLUSION

In conclusion, the growth analysis of mustard plants revealed significant results in their performance based on different treatments. While Sole mustard (T8) remained the best performer out of all, the treatment 60% Pruning + Apical cutting (T6) great potential as it often performed at par if not better than the sole crop. The treatments T4 and T5 also show great potential when it comes to growth parameters of mustard in agroforestry.

These findings shed light on the impact of specific treatments on the growth and development of mustard plants, providing valuable insights for agricultural practices and crop management.

The yield analysis further indicated variations in biological yield, grain yield, and straw yield, with Sole mustard (T8) showing the highest values. Additionally, the grain to straw ratio and the harvest index were found to be consistent across different treatments. Biometric analysis of neem trees has revealed variations in various parameters, with 60% Pruning + Apical cutting (T6) frequently yielding the most favourable results. The tallest trees and the highest girth, diameter, volume, above ground biomass, below ground biomass, total biomass, above ground carbon, below ground carbon and total carbon were frequently associated with this treatment. These findings contribute to the understanding of agroforestry management and the potential impact of pruning and apical cutting on neem tree growth.

Thus, the pruning regimes followed in this study have proven successful as they have significantly closed the gap in the growth and yield parameters between sole cropping and neem-based agroforestry while also influencing the desirable growth and yield of neem trees.

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